BIO DATA

Name : Dr. P.Prem

Designation : Associate Professor-III

Department : EEE

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1. Educational Qualification

Degree	Branch / Specialization	Institute/University	Class	Year of Passing
B.E.	Electrical and Electronics Engineering	CSI Institute of Technology, Thovalai	**First Class**	2006
M.E	Power System Engineering	Thiagarajar College of Engineering, Madurai	**First Class with Distinction.** .	2008
Ph.D	Electrical Engineering	Anna University, Chennai		2018

2. Professional/ Industry Experience:

S.No	Designation	Institution/ Organization	Period	
			From	To
1	Associate Professor	Ramco Institute of Technology, Rajapalayam	15.04.2024	Till date
2	Associate Professor	St.Thomas' College of Engineering & Technology, Kolkata	21.10.21	09.04.2024
3	Manager-Operations & Business Development	Switch Gear Electromechanical LLP	03.03.2021	08.10.2021

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4	Associate Professor	Bannari Amman Institute of Technology, Sathymangalam	01.11.2019	22.12.2020
5	Assistant Professor – Level II	Bannari Amman Institute of Technology, Sathymangalam	01.07.2013	31.10.2019
6	Assistant Professor	Bannari Amman Institute of Technology, Sathymangalam	01.01.2011	30.06.2013
7	Lecturer	Bannari Amman Institute of Technology, Sathymangalam	18.06.2008	31.12.2010

3. Research Interest : Multilevel inverters, Industrial power system analysis

4 Publications

4.1 JOURNAL PUBLICATIONS

1. Ravindran S, S., R., Koothu Kesavan, P., Banjerdpongchai, D., **Prem P**, Modelling, implementation and analysis of double-side slotted axial flux PMGs suitable to small-scale wind energy conversion systems. Sci Rep 15, 20384 (2025). https://doi.org/10.1038/s41598-025-08716-6

2. Khan A R, Sathik M J, Gopinath N P, **Prem P,** Alamri F S, Bahaj Saeed Ali, "A 7L and 11L High Step-Up SCMLI Topology with Reduced Component Voltage Stress", DOI: 10.1109/ACCESS.2023.3333363 IEEE Access.

3. Algadair, Badr Saleh A., Mahmoud F. Elmorshedy, Jagabar Sathik Mohamed Ali, Dhafer Almakhles, and **P Prem**. "High Step-Up Thirteen Level Switched-Capacitor Inverter Topology for HFAC Power System." Processes 11, no. 3 (2023): 848.

4. Rawa, Muhyaddin, **Prem P**, Jagabar Sathik Mohamed Ali, Marif Daula Siddique, Saad Mekhilef, Addy Wahyudie, Mehdi Seyedmahmoudian, and Alex Stojcevski. "A new multilevel inverter topology with reduced dc sources." Energies 14, no. 15 (2021): 4709.

5. **Prem, P,** Jagabar Sathik, and V. Suresh. "A new generalized cross-connected switched capacitor multilevel inverter topology with the high gain output voltage for single phase solar PV unit." Journal of Engineering Research 9, no. 4A (2021)

6. Suresh V, Senthil Kumar R, **Prem P** & Jagabar Sathik M "Design of Nine Step Switched Capacitor Multilevel Inverter and Its Cascaded Extension". **International Journal of Circuit Theory and Applications**, 2021; 1– 20. https://doi.org/10.1002/cta.2926

7. Alagu, M, **Ponnusamy Prem**, Pandarinathan, S, Mohamed Ali, JS. Performance improvement of solar PV power conversion system through low duty cycle DC-DC converter. **International Journal of Circuit Theory and Applications**, 2020; 1– 16. https://doi.org/10.1002/cta.2918.

8. Maheswari K.T l, Bharani Kumar R & **Prem Ponnusamy** (2020) Performance Analysis of Dual Space Vector Modulation Technique-Based Quasi Z-Source Direct Matrix Converter, **Electric Power Components and Systems**, 48:11, 1185-1196, DOI: 10.1080/15325008.2020.1834018

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9. **Prem P. et al.** "A New Multilevel Inverter Topology With Reduced Power Components for Domestic Solar PV Applications," in **IEEE Access**, vol. 8, pp. 187483-187497, 2020, doi: 10.1109/ACCESS.2020.3030721.

10. **Prem Ponnusamy**, Suresh Velliangiri & Jagabar Sathik Mohamed Ali (2020) A Hybrid Switched Capacitor Multi-Level Inverter with High Voltage Gain and Self-Voltage Balancing Ability, **Electric Power Components and Systems**, 48:8, 755-768, DOI: 10.1080/15325008.2020.1821832

11. **P.Prem**, P.Sivaraman, J.S.Sakthi Suriya Raj, M.Jagabar Sathik, Dhafer Almakhles "Fast charging converter and control algorithms for Solar PV battery and electrical grid integrated Electric Vehicle Charging Station", **Automatika: Journal for Control, Measurement, Electronics, Computing and Communications**. DOI:10.1080/00051144.2020.1810506

12. S. Bhuvaneswari, P. Sivaraman, A.Matheswaran, **P.Prem**, "Performance Analysis of Radial Flux and Axial Flux Permanent Magnet Generators for Low- Speed Wind Turbine Applications", **International Journal of Applied Electromagnetics and Mechanics**, 1 Jan. 2020 : 1 – 19.

13. **P.Prem,** Vidyasagar Sugavanam, Ahamed Ibrahim Abubakar, Jagabar Sathik Mohamed Ali, Boopathi C Sengodan, Vijayakumar Krishnasamy Sanjeevikumar Padmanaban," A novel cross-connected multilevel inverter topology for higher number of voltage levels with reduced switch count", **International Transaction on Electrical Energy Systems**, Vol.30, Issue:6, e12381.

14. **P.Prem**, Jagabar Sathik, Sivaraman P., Matheswaran A. & Shady H. E. Abdel Aleem "A New Asymmetric Dual Source Multilevel Inverter Topology With Reduced Power Switches", **The Journal of Chinese Institute of Engineers**, Vol.42, Issue:5, April 2019, pp.460-472.

15. P.Sivaraman, **P.Prem** "PR controller design and stability analysis of single stage T-source inverter based solar PV system", **The Journal of Chinese Institute of Engineers**,Vol.40, Issue:3, March 2017, pp.235-245

4.2 Book/Book chapter publications :

1. Matheswaran, A., **P. Prem**, C. Ganesh Babu, and K. Lakshmi. "Opportunities of Translating Mobile Base Transceiver Station (BTS) for EV Charging Through Energy Management Systems in DC Microgrid." Smart Grids for Smart Cities Volume 1 (2023): 15-40.
2. **Prem.P**, Maheswari.K.T, M.Jagabar Sathik, Chapter 10- Multilevel Converters and Applications, Editor(s): Dr.Mahajan Sagar Bhaskar, Dr.P.Sanjeevikumar Padmanabhan, Dr.Nikita Gupta, Dr.Dhafer J Almakhles, DC Microgrid-Advances, Challenges, and Applications , Scrivener Wiley Publications, 2021 (Under Press)

3. Maheswari.K.T, **Prem.P**, M.Jagabar Sathik, Chapter 17- Power Electronics: Technology For Wind Turbines, Editor(s): Dr.Mahajan Sagar Bhaskar, Dr.P.Sanjeevikumar Padmanabhan, Power Electronics For Green Energy Conversion, Scrivener Wiley Publications, 2021 (Under Press)

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4. J. Senthil Kumar, **P. Prem**, M. Jagabar Sathik, Chapter 7 - Evaluation of DG impacts on distribution networks, Editor(s): Ahmed F. Zobaa, Shady H.E. Abdel Aleem, Uncertainties in Modern Power Systems, Academic Press,2021, Pages 195-209,ISBN 9780128204917,https://doi.org/10.1016/B978-0-12-820491-7.00007-4.

5. N. Rishikesh, **P. Prem**, M. Jagabar Sathik, Dhafer Almakhles, Chapter 15 - A case study with power quality analysis on building integrated PV (BIPV) system, Editor(s): Ahmed F. Zobaa, Shady H.E. Abdel Aleem, Uncertainties in Modern Power Systems, Academic Press, 2021, Pages 541-562, ISBN 9780128204917, https://doi.org/10.1016/B978-0-12-820491-7.00015-3.

5 Funded Projects

S.No.	Title of the project	Funding Agency	Duration	Amount	Status
1	Aakash Equipment funding	MHRD	2012	1,00,000	Completed
2	Establishment of A-View infrastructure to facilitate students to interact with Professors from IITs	MHRD	2015	2,80,000	Completed
3	A one-week STTP on “Electric Power Systems”	MHRD	2017	85,000	Completed
4	Modernization of Electrical Machines Laboratory	AICTE-MODROBS	2019	13,88,000	Completed

6 Ph.D. guidance

Sl.No	Scholar name	University	Topic	Status
NIL				

7 Short term Courses / Seminars /FDP/Workshop/ConferenceAttended:

Sl.No	Title f the event	Organized by	Date	
			From	To
1.	Navigating the Digital Landscape: Next-Gen Communication techniques for Teaching Pedagogy and Research Writing	ATAL FDP, Knowledge Institute of Technology, Salem	20.01.2025	25.01.2025
2.	One Week FDP on National Education Policy	NITTTR, Kolkata	17.07.2023	21.07.2023

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3.	One Week FDP on Power System Analysis	College of Engineering, Guindy	27.05.2019	31.05.2019
4.	One Week ISTE STTP for Coordinators on Electric Power System	Indian Institute of Technology, Kharagpur	01.05.2017	05.05.2017
5.	Energizing South -2016: Smart-Reliable-Sustainable Power	Confederation of Indian Industries (CII)	22.08.2016	23.08.2016
6.	Grounding Practices	Central Power Research Institute, Bengaluru	28.01.2016	29.01.2016
7.	FDP on Power Quality	Anna University, Chennai Sponsored FDTP	18.12.2013	24.12.2013
8.	Challenges of Power System Planning in Restructured Indian Electricity market	Pondicherry Engineering College	13.12.2013	14.12.2013
9.	Current Issues and Potential Research in Power Industry	Coimbatore Institute of Technology, Coimbatore and Victoria University, Melbourne	15.12.2012	
10.	Coordinators' workshop for Aakash coordinators	IIT Bombay	05.11.2012	
11.	AICTE SDP on Soft Computing Techniques in electrical systems and control	Government college of Technology, Coimbatore	09.05.2012	22.05.2012
12.	Power Quality & Energy Management – Challenges	VIT, Vellore	18.10.2011	
13.	One day workshop on "Energy Conservation & Renewable Energy"	Confederation of Indian Industries & Tamilnadu Electricity Consumers Association	08.08.2011	

8. Short term Courses / Seminars /FDP/Workshop/Conference organized:

Sl.No	Title of the event	Funded by	Date	
			From	To
1.	FDP on Competency based Engineering Education	Ramco Institute of Technology	10.02.25 to	15.02.2025

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2.	Use of ICT in Education for Online and Blended Learning	Funded by SAP India PVT Ltd through CSR grants	Week ends of May-July 2016	
3.	One-day work shop on "Illumination Systems"	Bannari Amman Institute of Technology	10.09.2016	
4.	Two-Week ISTE workshop on "Signals & Systems"	Jointly with IIT Kharagpur and MHRD, India	02.01.2014	12.01.2014
5.	Research Methods in Educational Technology	IIT Bombay and MHRD	02.02.2013	09.02.2013
6.	Aakash for Education	IIT Bombay and MHRD	10.11.2012	11.11.2012
7.	Energy Audit - A Practical Exposure	Bannari Amman Institute of Technology	03.03.2011	05.03.2011
8.	Modern Drives in Industrial Applications	Bannari Amman Institute of Technology	14.09.2010	15.09.2010
9.	National Conference on Cutting Edge Technologies on Power Conversion and Industrial Drives	*CSIR, Larsen & Toubro, Indian Value Engineering Society*	12.03.2010	1303.2010

9. Online Courses

S. No	Course title	University	Date
1.	Patent Drafting for Beginners	IIT, Chennai	March 2024
2.	Road Map for Patent Creation	IIT, Kharaghpur	March 2024
3.	Programming Data Structures and Algorithms using Python	Chennai Mathematical Institute	September 2025

10. Patent filed/Copyright filed/awarded:

Sl.No	Title of Invention	Date of Filing / Publication	Patent No / Application No	Status of Indian Patent
1	Antiviral respirator integrated helmet	02/09/2020	333454-001	**Granted**
2	Smart Coin Collecting Carom	17/10/2020	334397-001	**Granted**
3	Detachable electric motor drive for Wheelchair	21.08.2024	202441063082	**Filed**

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4	Machine Learning Based Anomaly Detection And Monitoring In Battery Energy Storage Systems	24.06.2025	202541060046	**Filed**

11. Membership in Professional societies : ISTE

12. Awards, Recognition & Achievements:

1. **Headed the Department of Electrical Engineering** at St.Thomas' College of Engineering & Technology, Kolkata from 01.07.2022 to 09.04.2024

2. **Project Manager** represented Switch Gear Electromechanical at **Chennai Airport** Expansion Project

3. **Headed the Department of EEE** in Bannari Amman Institute of Technology during 30.08.2018 to 01.11.2020 (2 years, 3 Months)

4. **Recipient of IEEE Publication award** from IEEE Madras section for the year 2020

5. Completed an **M.B.A degree in Education management** through Distance Education mode at Alagappa University, Tamilnadu during 2008-2010 and secured first class

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